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(54) **MULTIPOLE COILS**

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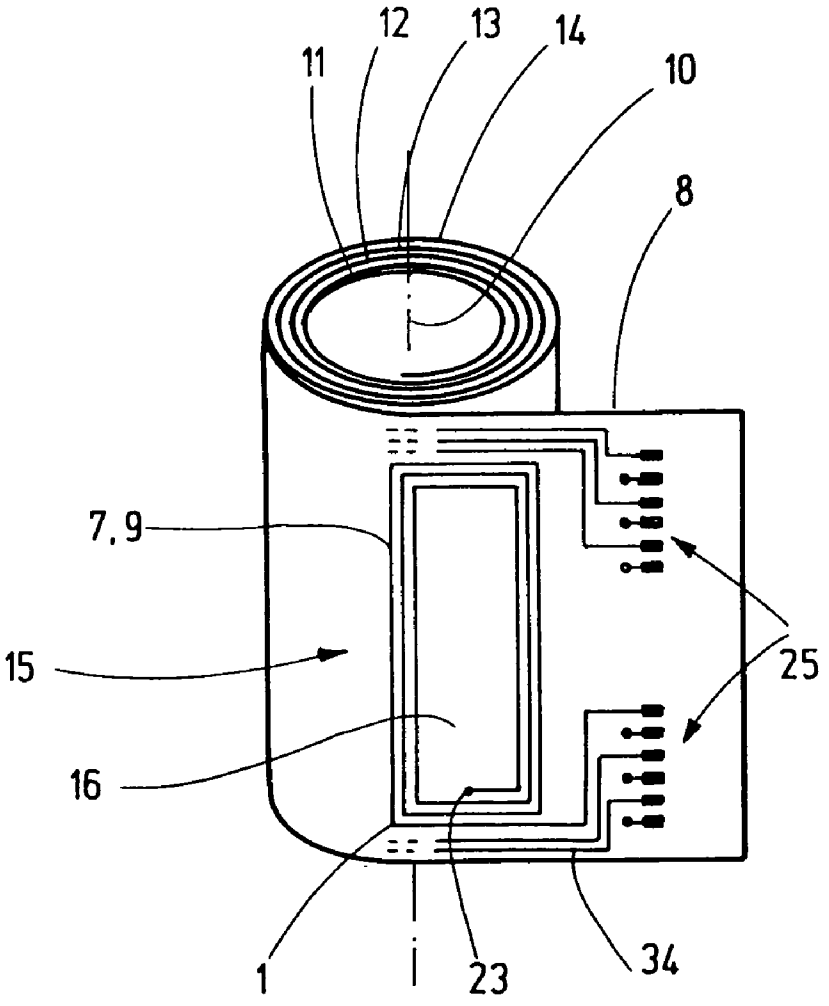
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(57) **ABSTRACT**

Multipole coils (**1**, **2**, **3**, **4**, **5**, **6**) comprise at least two coils (**1**, **2**) which are disposed to concentrically enclose an imaginary axis (**10**). Multipole coils (**1**, **2**, **3**, **4**, **5**, **6**) of this type are designed in such a fashion that effective fields can be generated in the area of an imaginary axis (**10**) when little installation space is available, and the multipole coils can be reproducibly manufactured with high precision. This is achieved in that, for each coil (**1**, **2**, **3**, **4**, **5**, **6**), at least one winding (**7**) is disposed on a flexible printed circuit board (**8**) through disposed strip conductors (**9**), and the printed circuit board (**8**) is rolled in at least one printed circuit board layer (**11**, **12**, **13**, **14**).



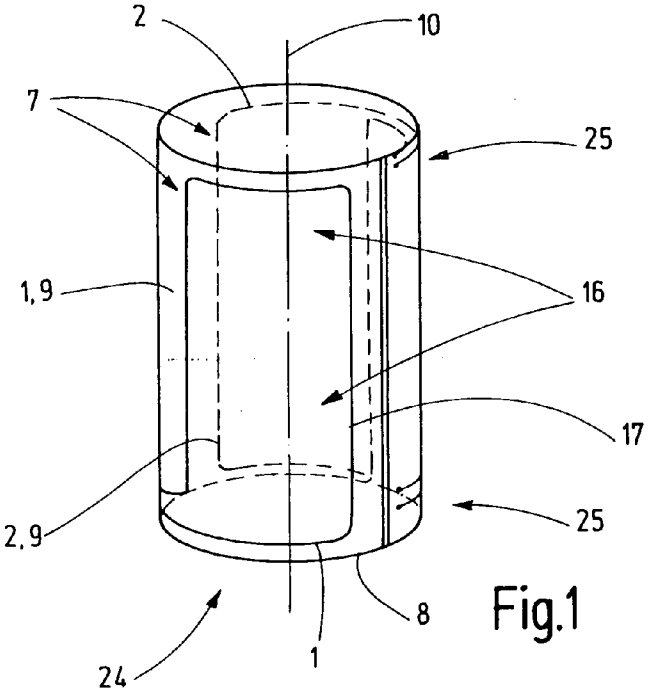


Fig.1

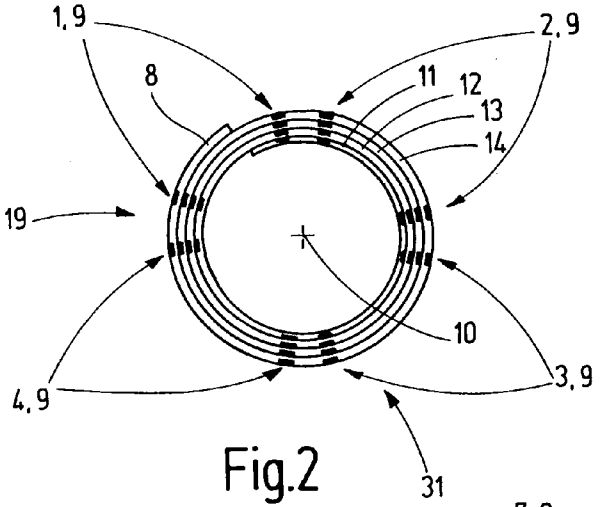


Fig.2

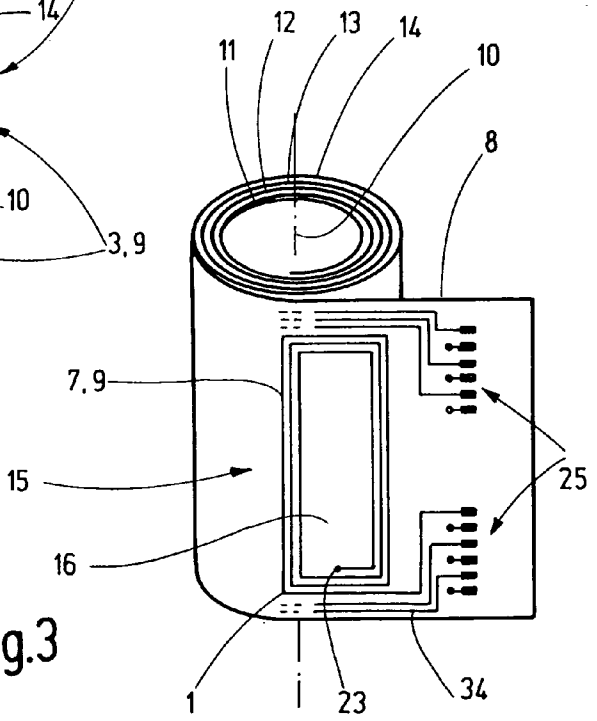


Fig.3

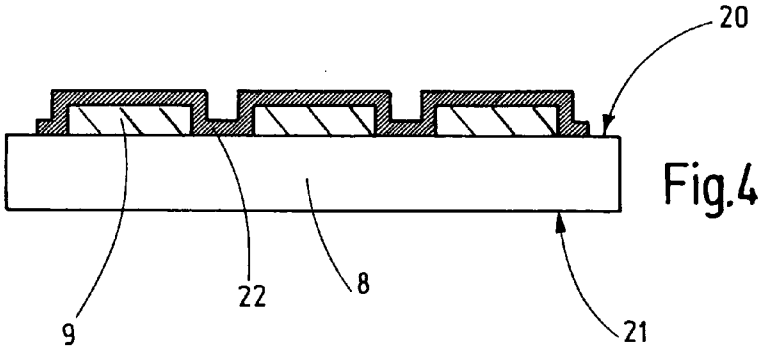


Fig.4

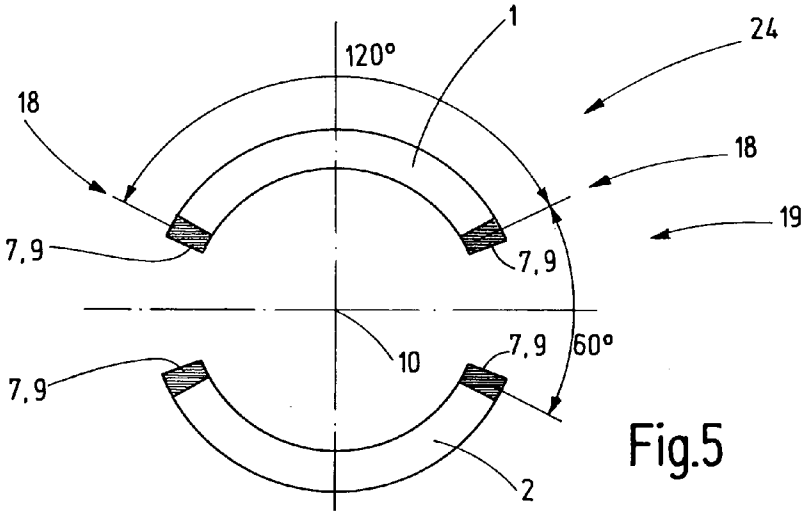


Fig.5

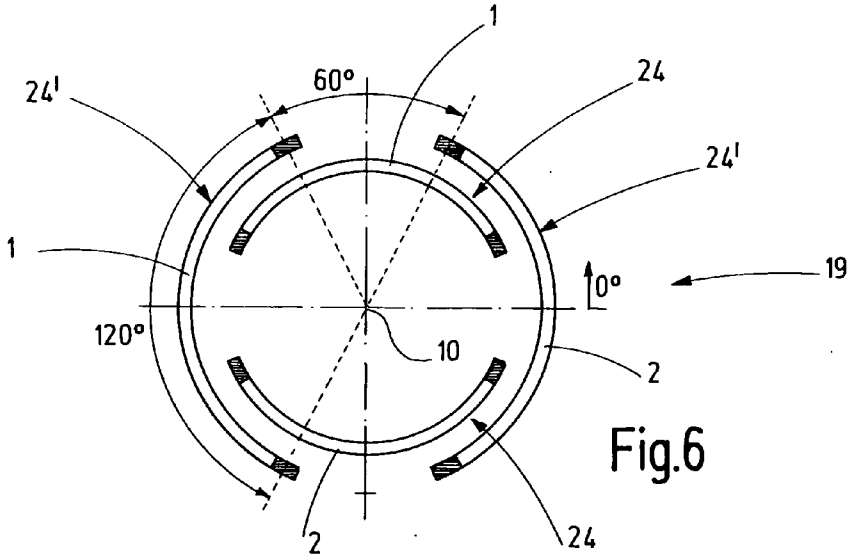


Fig.6

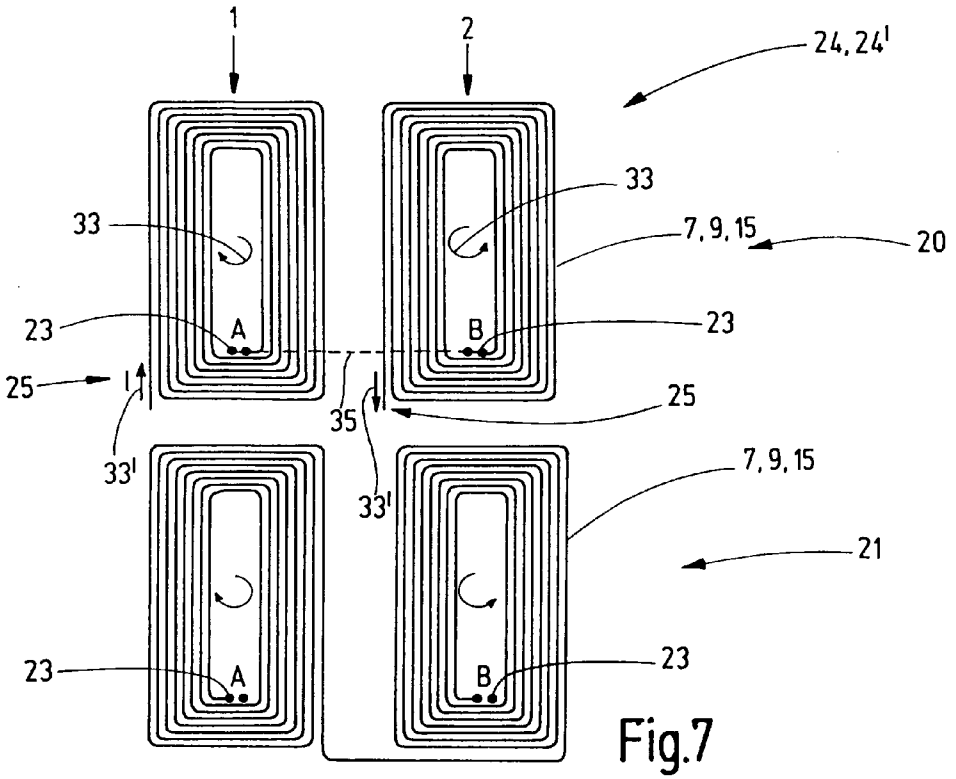

1
2
24, 24'
33
33
7, 9, 15
20
23
A
B
23
25
I
I
25
35
33'
33'
7, 9, 15
21
23
A
B
23

Fig.7

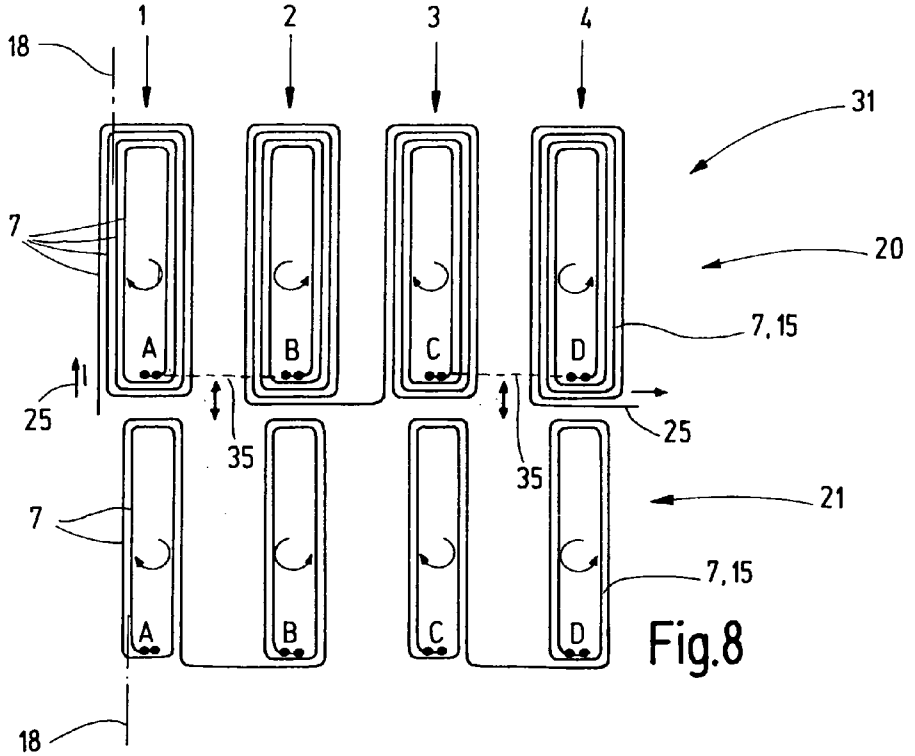

1
2
3
4
18
31
7
20
A
B
C
D
7, 15
25
I
25
35
35
21
7
7, 15
A
B
C
D
18

Fig.8

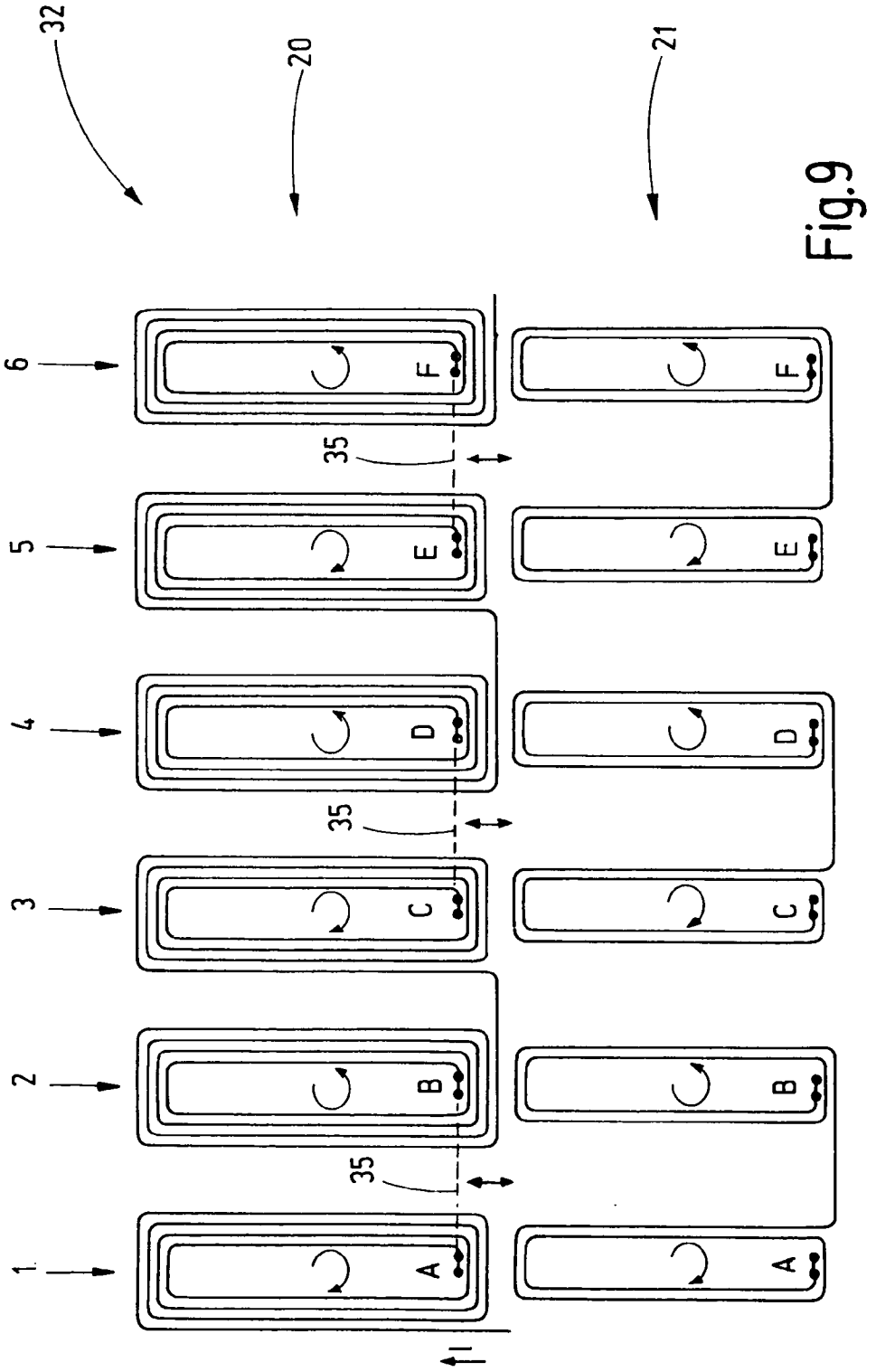


Fig.9

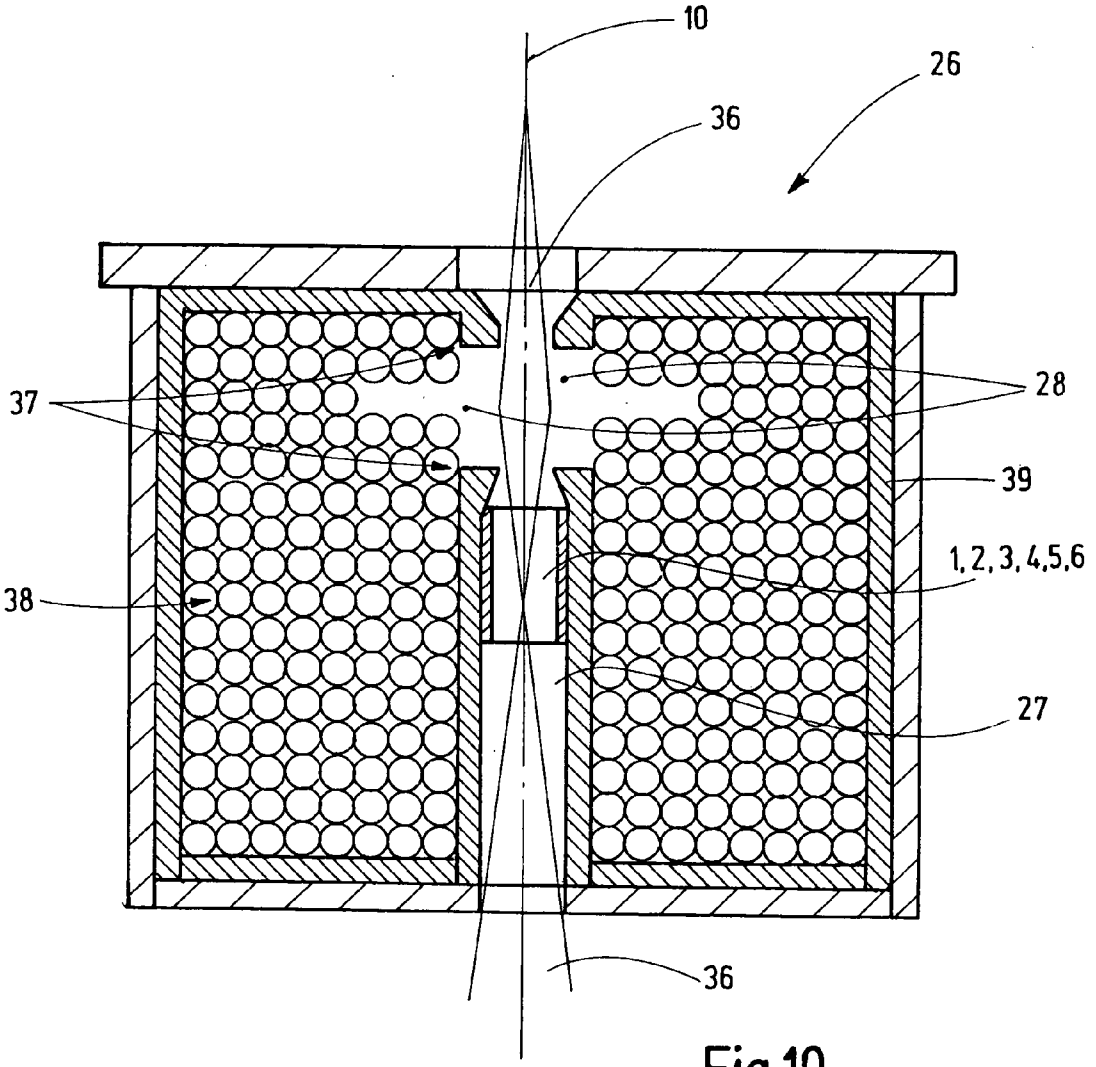


Fig.10

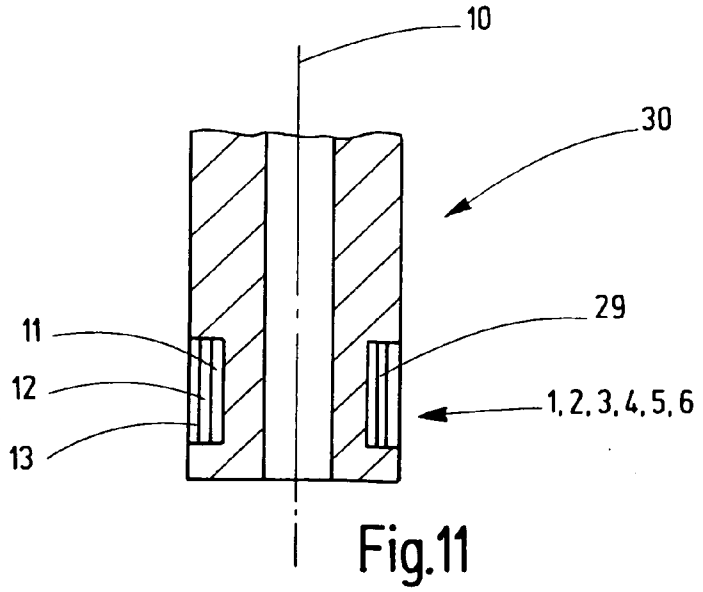


Fig.11

MULTIPOLE COILS

[0001] This application claims Paris Convention priority of DE 10 2007 045 874.8 filed Sep. 25, 2007 the complete disclosure of which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] The invention concerns multipole coils comprising at least two coils which are disposed to concentrically enclose an imaginary axis. Multipole coils of this type are used to generate magnetic fields which are disposed concentrically about an imaginary axis. A dipole, a quadrupole, a hexapole, an octupole etc. can e.g. be formed in this fashion.

[0003] The main field of application is the influence of particle beams, i.e. ion or electron beams in particle optics. Since 1947, when O. Scherzer showed that the optical path of electron lenses can be corrected by non-rotationally symmetrical lenses (O. Scherzer: "Sphärische und chromatische Korrektur von Elektronen-Linsen" (spherical and chromatic correction of electron lenses), OPTIK, DE, JENA, 1947, pages 114-132, XP002090897, ISSN: 0863-0259) this type of correction has been generally used (David C. Joy "The aberration corrected SEM"—http://www.ccl.nist.gov/812/conference/2005_Talks/Joy.pdf or David B. Williams and C. Barry Carter "Transmission Electron Microscopy", ISBN 0-306-45324-X, in particular page 93). Dipoles, quadrupoles, hexapoles and octupoles can be generated, in which the electromagnetic coils are disposed in such a fashion that the south and north poles are alternately disposed along the periphery. This enables influencing an electron or ion beam that passes through the coil arrangement in the area of the axis (optical axis) in order to correct aberrations.

[0004] Other fields of application are naturally also feasible, such as e.g. deflection of particle beams, e.g. for scanning electron microscopes or corresponding deflection of particle beams for other technical purposes, e.g. in order to process surfaces.

[0005] The multipoles of prior art correspond to the representation of FIG. **6.8**(D) of David B. Williams and C. Barry Carter, page 93. These are webs that face towards the optical axis and are wrapped with conducting wires. Multipoles of this type require a large amount of space, comprise fields which are ineffective with respect to the area of the axis due to the large distance from the axis, and cannot be manufactured in an exactly reproducible fashion, since deviations cannot be prevented during winding of the wires both manually as well as mechanically.

[0006] Moreover, mst/infobörse by the "Bundesministerium für Bildung und Forschung" (Federal Ministry for Education and Research) "Mikrosystemtechnische Lösungen für stromtragfähige induktive Bauelemente (Mikrosyst)" (micro-system technical solutions for current-carrying inductive components) (www.mst-innovation.de) discloses the production of coils by arranging coil windings on printed circuit boards and subsequently stacking them e.g. by folding. It is also possible to join several coils of this type to form multipoles, but a reproducible spatially concentric arrangement of several coils about an imaginary axis is not suggested. Each coil would have to be positioned with great effort, the position would have to be examined by measurement techniques and possibly corrected.

[0007] It is therefore the underlying purpose of the invention to provide multipole coils of the above-mentioned type which produce effective fields in the area of an imaginary axis and require little space, and to manufacture the multipole coils with great accuracy in a reproducible fashion.

SUMMARY OF THE INVENTION

[0008] This object is achieved in accordance with the invention in that, for each coil, at least one winding is disposed on a flexible printed circuit board through disposed strip conductors, and the printed circuit board is rolled into at least one printed circuit board layer.

[0009] The invention enables the production of multipoles that require a minimum amount of space. Multipoles having an uneven pole number, e.g. three, five, or seven poles, may also be produced in addition to the above-mentioned multipoles.

[0010] Multipoles of conventional construction which are disposed about a beam tube area of 6 to 10 mm in particle optics, have an outer diameter which is approximately 2 cm larger than the beam tube area. The inventive multipole coils require diameter enlargement of less than 1 cm. In most cases, a wall thickness of the rolled printed circuit boards of less than 3 mm is sufficient, even when they comprise several layers. These space reductions are naturally also advantageous for other fields of application.

[0011] The printed circuit board is thereby generally rolled into a body having a cylindrical surface. The shape could also be different. The roll could also have a polygonal cross-section instead of a circular ring-shaped cross-section, wherein the corners would naturally have to be rounded in correspondence with the deformability of the printed circuit board.

[0012] The reduction in installation space is also advantageous in that the coils are positioned closer to the location of use of the generated fields, and therefore produce fields of higher precision. The number of windings that are required for generating the field is also smaller compared to prior art, such that considerably stronger fields can be generated at the location of use with the same current strength. The field that can be generated using the same current strength is approximately four times stronger. Alternatively, it is also possible to use a correspondingly smaller current strength to obtain the same field.

[0013] Manufacture can be realized using standard printed circuit board technology, wherein the lithographic or a similar conventional production method always ensures reproducible, exact positioning of the windings of the multipole coils. This enables economic mass production with very high precision, and the expense for manual or mechanical winding around webs is avoided. The latter is, in fact, often difficult, since the wires must be wound around the webs. When two coil bodies are disposed on top of each other in the axial direction, the web of one coil body obstructs winding of the other, and even manual production of a winding is very difficult.

[0014] Due to the higher production accuracy, the produced examples are almost identical, and the high calibration effort that was previously required for compensating the production differences through corresponding variations in the current load, is eliminated. Manufacture is thereby considerably facilitated and the production costs substantially reduced, in particular, for large quantities. The capacitive properties of the excitation coils can be precisely influenced, e.g. mini-

mized, through corresponding selection of the conductor separations or dielectric properties of the printed circuit board.

[0015] One advantage, which is very important, in particular, in the field of particle optics, consists in that very large windows can be produced which are surrounded by windings, since the coils of the individual poles can tightly abut each other on the printed circuit boards, or even overlap when disposed in different layers. In consequence thereof, pure fields of the respective multipole can be generated and no other multipole fields are superimposed, which is the case with wound coils, in particular, when these coils have larger separations. One example of such overlapping are dipoles that are superposed with a hexapole field due to the conventional coil geometry.

[0016] The conventional multipoles also had unavoidable angular errors with respect to pole division, since the webs and windings could not be produced with sufficient precision. The invention eliminates this problem, since rolling of the printed circuit boards enables much more precise positioning.

[0017] Another advantage is the improved heat dissipation, since gaps containing air form between the wires during production of the windings. Heat dissipation is improved by exactly disposing the printed circuit boards on top of each other.

[0018] In addition to increasing the current, the fields generated by the coils can also be reinforced by generating several windings per coil through spiral arrangement or through several printed circuit board layers generated during rolling. A combination of both is naturally also possible, which provides excellent field generation efficiency. When several layers are provided, measures must be taken to position the windings that interact as coils, or the spiral arrangements of windings with great precision with respect to each other.

[0019] The windings of the coils of the different printed circuit board layers suitably extend in such a fashion that the winding centers are always disposed on radii in a plane that extends perpendicularly to the axis. This widens the window formed by the coil windings, from the inside to the outside in correspondence with the increasing radius. A hexapole comprises six coils, which provides e.g. an opening angle of approximately 60° each with respect to the axis. This coil design cannot be realized with conventional technology. It is advantageous in that fields of considerably higher correctness of the different multipoles can be generated, since the separation between individual coils can be minimized. This is important, in particular, for a large number of poles. Moreover, reproduction of the generated fields is considerably facilitated through the exact positioning of the windings.

[0020] The coils are thereby preferentially designed to form substantially rectangular windows, wherein the long sides of the windows suitably extend in the axial direction and the short sides in the radial direction, thereby generating more effective fields.

[0021] When the windings are spirally arranged in several printed circuit board layers, the outer spiral arrangements suitably comprise more windings than the inner ones. This can be realized on the one hand by providing more space in this area, which is advantageous in that with increasing distance from the axis, the field excitation increases, such that a homogeneous field of maximum strength is provided in the area of the axis. The above-mentioned winding centers are thereby located in the center of the neighboring strip conductors of the spiral arrangement to also enable the above-mentioned outward window enlargement with respect thereto. The field excitation which is stronger to the outside, could naturally also be realized using larger conductor cross-sections and larger current loads, which would, however, generally require separate current supply lines.

[0022] Field excitation can be further improved by doubling the number of windings per printed circuit board layer by providing the front and rear sides of the printed circuit board with strip conductors that form windings. Of course, the strip conductors of the printed circuit board layers must then be separated by an insulating layer.

[0023] The terminals and connections of the different windings that belong to one coil can be realized via strip conductors on the printed circuit boards, which would be disadvantageous with respect to space requirements and generation of interfering fields. For this reason, it is proposed to connect different planes of strip conductors using through connections that pass through the printed circuit board. In this fashion, all windings that belong to one coil can be connected to each other in a simple fashion, and only the different coils of a multipole must then be connected to each other via paths on the printed circuit boards or paths in connection with through connections.

[0024] The strip conductors should have a high current carrying capacity to generate fields of corresponding strength. For this reason, it is proposed to increase the cross-sectional surface of the strip conductors through galvanic reinforcement. The strip conductors must thereby consist of material having good conducting properties, normally copper.

[0025] The generated fields are more precise, the exacter the positioning of the windings of the respective coils. In order to obtain such positioning for generating several printed circuit board layers during rolling, the invention proposes to provide adjustment markings for these positions on the printed circuit boards. The adjustment markings may be bores through which a mandrel is inserted. The printed circuit boards are subsequently suitably glued to each other on a large area in order to secure the exact position and at the same time provide good heat dissipation.

[0026] In order to further improve this heat dissipation, thermally conducting materials may be used which carry the thermal flow to the outside. The glue that is used for gluing the printed circuit boards may be used for this purpose if it has good heat conducting properties. Additional materials may naturally also be provided for this purpose.

[0027] Additional electric or electronic components with e.g. measuring and regulation functions may also be provided on the printed circuit board.

[0028] The coils of a core-less basic construction may naturally also be provided with a hard or soft magnetic core, or soft or hard-magnetic materials may be used at any location to influence the magnetic flux. Cavities in the printed circuit board, which can be produced e.g. through openings in the window area of the coils, may be utilized for this purpose. Magnetic material may also be disposed on the outer side of the printed circuit board, which connects the coil cores in the form of a yoke.

[0029] Multipole coils of this type can also be used for other applications in addition to the above-mentioned purpose of use of the inventive multipole coils.

[0030] One further possible application would be the detection of magnetic alternating fields, i.e. dipole or quadrupole

fields etc., which induce currents in the corresponding multipole coils and whose position and intensity can thereby be detected.

[0031] As mentioned above, particle optics is a particularly important field of use of such multipole coils, in particular electron optics. Towards this end, the multipole coils are suitably disposed around a vacuum tube that contains the optical axis. Its purpose is mainly to correct aberrations, as mentioned above, in particular, to correct minor aberrations caused by mechanical inaccuracies of the lenses. This is important, in particular, in electron microscopy, which requires high image resolution that cannot be realized without such corrections. There are also further fields of use. Multipoles of this type can e.g. also be used as scanning coils in a scanning electron microscope, wherein, in this case, two dipoles are suitably provided, one being used for deflection in the x direction and the other for deflection in the y direction.

[0032] Dipoles are advantageously designed in such a fashion that the centers of the windings of each coil are disposed on radii that form an angle of approximately 120° with respect to the axis. This is advantageous in that a relatively pure dipole field is generated, and formation of a dipole field superposed by a hexapole field is prevented, in contrast to prior art.

[0033] The inventive arrangement is particularly advantageous in that several identical or also different multipoles can be disposed on a printed circuit board. As mentioned above, two dipoles, or even dipoles, quadrupoles, hexapoles, octupoles etc. may be disposed on a printed circuit board, in order to generate combined multipole fields, wherein each multipole is suitably separately driven, thereby permitting correction of aberrations of different orders with exactly the same device. Towards this end, the multipoles must have separate connections for the current load. Multipoles may naturally also be directly connected to each other or by introducing further components in order to superpose different multipole fields that must be positioned in a certain relationship with respect to each other, in order to correspondingly influence the particle beam.

[0034] In one particularly suitable example of an arrangement of several multipoles, a dipole and a further dipole, rotated through 90° relative to the first, are disposed on the same printed circuit board. When these two dipoles are disposed on different printed circuit board layers, both can extend over an angle of 120° per coil, as mentioned above, since both abutment as well as overlappings are possible. This yields fields of particular precision and efficiency, which cannot be achieved at all with conventional coils. In the above-mentioned combination or in other combinations, several multipoles may be disposed e.g. to be distributed over several printed circuit board layers, or several multipoles may be disposed on top of each other in the axial direction. Combinations of both arrangements are also possible. In any case, the exact mutual allocation of these multipoles is advantageous, and no significant angular error occurs during allocation. Another simple variant is to roll the printed circuit board only in several layers in order to obtain an arrangement of this type.

[0035] Such combinations of several different multipoles are preferably used in particle optics for correcting various aberrations of different orders. This permits correction of almost any aberration using one single correction element.

[0036] In order to prevent a further increase in construction height of the particle optics through introduction of the multipole coils, the printed circuit board may be arranged in the beam passage opening of a lens. A multipole coil of this type can naturally also correct aberrations of upstream or downstream lenses.

[0037] In a magnetic lens, the coils are preferably arranged between the magnetic gap and the lens boundary, wherein corresponding widening of the beam passage opening, e.g. through a recess is possible, if required in view of space requirements, such that the multipole coils can be accommodated between the inner lens wall and the vacuum tube. The printed circuit board with multipoles can naturally also be inserted into other recesses of a particle optical element, wherein the printed circuit board may also simply be disposed into a depression during rolling, optionally in several layers.

[0038] The invention is advantageous compared to prior art due to the minimum space requirements. The accommodation of conventional, wound coils would require considerably larger recesses, e.g. in a magnetic lens. This would change the geometry of such a lens and have a negative effect on the generated magnetic field. The small installation space of the inventive multipole coils completely or largely eliminates such geometrical changes. A correction element can thereby be inserted with minimum disturbance of the lens field.

[0039] The invention is explained below with reference to embodiments shown in the drawing.

BRIEF DESCRIPTION OF THE DRAWING

[0040] FIG. **1** shows a simple embodiment of the invention to explain the basic principle;

[0041] FIG. **2** shows a sectional view of an embodiment with the printed circuit board being rolled into several printed circuit board layers;

[0042] FIG. **3** shows a 3-dimensional view of the printed circuit board being rolled in several layers;

[0043] FIG. **4** shows a section through a printed circuit board with strip conductors;

[0044] FIG. **5** shows a section through a dipole;

[0045] FIG. **6** shows a section through two dipoles which are disposed in different printed circuit board layers;

[0046] FIG. **7** shows an example of a coil arrangement of a dipole;

[0047] FIG. **8** shows an example of a coil arrangement of a quadrupole;

[0048] FIG. **9** shows an example of a coil arrangement of a hexapole;

[0049] FIG. **10** shows an arrangement of the inventive multipole coils in a lens; and

[0050] FIG. **11** in a recess of another particle-optical component.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0051] FIG. **1** shows a simple embodiment of the invention in order to explain the basic principle. Two coils **1** and **2**, which each consist of one winding **7**, are disposed on a printed circuit board **8** for generating a dipole **24**. Each winding **7** is disposed on the printed circuit board **8** in the form of a strip conductor **9**, and is loaded with current via connections **25**. The windings **7** form rectangular windows **16** with rounded edges, wherein the long sides **17** of the windows **16** are oriented along an imaginary axis **10**. The axis **10** is the optical axis when the multipole coils are used in the field of particle optics. The printed circuit board **8** is made from a

flexible material and rolled into a cylinder which is hollow inside, such than the two coils **1**, **2** concentrically enclose the axis **10**.

[0052] FIG. **2** shows a sectional view of one embodiment, in which the printed circuit board **8** is rolled in several printed circuit board layers **11**, **12**, **13**, **14**. These multipole coils **1**, **2**, **3**, **4** form a quadrupole **31**, since four coils **1**, **2**, **3**, **4** are concentrically disposed about an imaginary axis **10**. The section in a plane **19** perpendicular to the axis **10** thereby shows the cut strip conductors **9** which extend perpendicularly to the axis **10**. This illustrates that the strip conductors **9** of the windings **7** of the different printed circuit board layers **11**, **12**, **13**, **14** are precisely stacked in a radial direction. This is particularly important to obtain an exact coil geometry and thereby exact fields, in particular, when the multipole coils **1**, **2**, **3**, **4** are intended for correcting aberrations in particle optics. Gluing can secure exact positioning.

[0053] FIG. **3** shows a perspective view of a printed circuit board **8** rolled into several layers **11**, **12**, **13**, **14**. This example shows the design of the strip conductors **9** as windings **7** in a spiral arrangement **15**. Only one spiral arrangement **15** is shown as an example. They are, in fact, superposed like the windings **7** of FIG. **2**. The strip conductors **9** of the spiral arrangement **15** of the different printed circuit board layers **11**, **12**, **13**, **14** can be connected by through-connections **23**, such that all windings **7** of one coil are interconnected. The coils that belong to a multipole element are correspondingly interconnected again to alternately generate one south pole and one north pole of the respective magnetic fields. A rolled printed circuit board **8** of this type may thereby also contain coils **1**, **2**, **3**, **4**, **5**, **6** etc. of several multipoles. The supply lines **34** guided on the edges may serve for interconnecting the coils of one multipole or also for current loading of one or more multipoles at the connections **25**. Also shown are the rectangular windows **16** formed by the windings **7**.

[0054] FIG. **4** shows a section through a printed circuit board **8** with strip conductors **9**. The strip conductors **9** are thereby mounted to the front side **20** of the printed circuit board **8** and covered by an insulating layer **22**. Strip conductors **9** may accordingly also be disposed on the rear side **21** of the printed circuit board **8**. This is advantageous in that twice as many windings **7** or spiral arrangements **15** of windings **7** can be formed as there are printed circuit board layers **11**, **12**, **13**, **14**. In this case, the insulating layer **22** is required for electrically separating the strip conductors **9**. During winding of the printed circuit board **8**, a glue layer is additionally provided on the full surface, which secures the positions of the coils and thereby also improves dissipation of heat. For this reason, a connection without air gaps is advantageous.

[0055] FIG. **5** shows a section through a dipole **24**, in which the printed circuit board **8** is not shown. This figure shows that each coil **1** and **2** extends over an area of 120°, which is utilized to generate a very pure dipole field. In this view, the strip conductors **9** are cut in a plane **19** perpendicular to the axis **10**, and the winding centers **18** are illustrated which, in simple windings **7** like this, extend in the center of the strip conductors **9**. Rolling of the printed circuit board **8** (not shown) provides the strip conductors **9** with the illustrated circular arc shape, which is also advantageous for generating fields which shall act with optimum precision on the area of the axis **10**.

[0056] FIG. **6** shows a similar section in a plane **19** perpendicular to the axis **10**, with two dipoles **24** and **24'** disposed on different printed circuit board layers **11**, **12**. Only the strip conductors **9** of the windings **7** of the coils **1** and **2** of the respective dipoles **24** and **24'** are thereby illustrated but not the printed circuit board **8** with its layers **11** and **12**. It is rolled in two layers in correspondence with FIGS. **2** and **3**. This shows how the dipoles **24** and **24'** can be superposed through two printed circuit board layers **11**, **12**, thereby providing an arrangement which cannot be produced using conventional wound coils. The coils **1** and **2** of the respective dipoles **24** and **24'** can naturally also be designed as spiral arrangements **15** in order to reinforce the generated magnetic forces. It is thereby also possible to provide more than two printed circuit board layers **11**, **12** to ensure that the number of windings **7** of the two dipoles **24** and **24'** can be further increased.

[0057] FIG. **7** shows one example of a coil arrangement **1**, **2**, of a dipole **24** or **24'**, wherein the windings **7** are spirally arranged **15**, and wherein the coils **1** and **2** are formed from windings **7**, **15** on the front side **20** of a printed circuit board, shown in the upper area of FIG. **7**, and additional windings **7**, **15** on the rear side **21** of the printed circuit board, shown in the lower area of FIG. **7**. The printed circuit board **8** is not shown either. Only the strip conductors **9** mounted to the printed circuit board **8** of the front side **20** are shown in the upper part of the figure and the strip conductors **9** of the rear side **21** of the printed circuit board **8** are shown in the lower part of the figure. The through-connections **23** are connected to each other at A and B to produce an electric connection **35** between A and B in the upper part of the drawing, which is symbolically shown with dashed lines and corresponds to the connection via the windings **7** of the spiral arrangements **15**, shown in the lower part of the figure. The printed circuit board **8** is rolled in correspondence with FIG. **1**. The dipole **24** or **24'** must then only be loaded with current via the connections **25** to produce a current flow, as indicated by arrows **33** and **33'**. In this fashion, two coils **1**, **2** are available which generate reversely poled magnetic fields.

[0058] FIG. **8** shows one example of a coil arrangement with coils **1**, **2**, **3**, and **4**, which form a quadrupole **31**. The view corresponds to the above-described, wherein the interconnection is correspondingly varied, such that the connections **35** shown in the upper part exist only between the coils **1** and **2** and the coils **3** and **4** via the corresponding windings **7** on the rear side **21** of the printed circuit board, and the windings **7** of the front side **20** of the printed circuit board are connected directly between the coils **2** and **3**. In this fashion, the coils **1**, **2**, **3** and **4** are also interconnected in the quadrupole **31**, which results in that the north and south poles of the magnetic fields alternate. The printed circuit board **8** is naturally also rolled in this case, such that the coils **1**, **2**, **3**, **4** concentrically enclose the axis **10**.

[0059] This view also shows that the spiral arrangement **15** has more windings **7** on the front side **20** of the printed circuit board **8** than the spiral arrangement **15** on the rear side **21** of the printed circuit board **8**. In consequence thereof, the outer spiral arrangements **15** have more windings **7** than the inner when the printed circuit board **8** is rolled in such a fashion that its rear side **21** faces to the inside and its front side **20** faces to the outside. The dash-dotted lines show that the winding centers **18** of each spiral arrangement **15** are in the center of the windings **7** which form the respective spiral arrangements **15**. These winding centers **18** must be arranged in such a fashion that they all coincide with the radii in a plane **19** perpendicular to the axis **10**. This applies not only for the front side **20** and the rear side **21** of a printed circuit board **8** but also for several printed circuit board layers **11**, **12**, **13**, **14** etc.

[0060] FIG. **9** shows another example of a coil arrangement of a hexapole **32** with six coils **1**, **2**, **3**, **4**, **5**, **6**. The windings **7** are thereby also shown as spiral arrangements **15** on the front side **20** of the printed circuit board **8** (top) and the rear side **21** of the printed circuit board **8** (below), wherein the printed circuit board **8** itself is not shown. The double arrows also indicate that the electric connections **35** between A and B, C and D, E and F of the windings **7** of the front side **20** are realized via the windings **7** on the rear side **21** by providing through-connections at points A, B, C, D, E and F. Otherwise, the above-described applies.

[0061] FIG. **10** shows an arrangement of the inventive multipole coils **1**, **2**, **3**, **4**, **5**, **6** in a lens **26** which is designed as a magnetic lens. The enumeration of six coils is naturally only an example. Any number of multipole elements may be disposed, as mentioned above, even a combination of several multipole elements. They are preferably arranged below the magnetic gap **28** formed by the pole shoes **37**, in the beam passage opening **27** of the lens **26**. The housing of the lens **26** is shown in sectional view, such that the windings **38**, which are also shown in sectional view, can be seen in the sectional view of the iron circuit **39**. The behavior of the electron beam **36**, which is influenced by the lens **26**, is also shown. Since the inventive multipole coils **1**, **2**, **3**, **4**, **5**, **6** require very little installation space, they may either be disposed directly in the beam passage opening **27** of the lens **26** without impairing the electron beam **36**, or if more space is required, a recess may be provided in the beam passage opening **27** for inserting the printed circuit board **8** including multipoles.

[0062] FIG. **11** shows such a recess **29** in an arbitrary particle-optical component, e.g. a steel tube **30**. A recess **29** of this type may be disposed on the outer side (as shown) or on the inner side of the beam passage opening **27** in correspondence to the lens **26** of FIG. **10**. The printed circuit board layers **11**, **12**, **13** etc. may thereby be easily inserted into any recess **29**. The recess **29** need not be designed in such a fashion that the rolled printed circuit board **8** can be mounted by sliding it on.

[0063] The drawings naturally only symbolically show a few embodiments. Many variations are feasible. The windings **7** could also be disposed in circles or ovals. The rectangular shapes could also have a different position, or much more complex arrangements could be provided by disposing many multipoles in many printed conductor board layers, which are used to correct a plurality of aberrations in particle optics or can also be used for other purposes. Depending on the purpose of use, the different coils must, of course, be correspondingly interconnected, wherein even separate control of individual coils would be feasible or also an interconnection of several multipole elements, possibly also by introducing further electric or electronic components. The drawing is only designed to give an exemplary idea of the invention.

LIST OF REFERENCE NUMERALS

[0064] **1**,**2**,**3**,**4**,**5**,**6** multipole coils or coils
[0065] **7** windings
[0066] **8** printed circuit board
[0067] **9** strip conductors
[0068] **10** axis, imaginary (e.g. optical axis)
[0069] **11**,**12**,**13**,**14** printed circuit board layers
[0070] **15** spiral arrangement
[0071] **16** rectangular window
[0072] **17** long sides of the windows
[0073] **18** winding centers
[0074] **19** plane extending perpendicularly to the axis
[0075] **20** front side of the printed circuit board
[0076] **21** rear side of the printed circuit board
[0077] **22** insulating layer
[0078] **23** through-connections
[0079] **24** dipole
[0080] **24'** further dipole
[0081] **25** connections for current load
[0082] **26** lens
[0083] **27** beam passage opening of a lens
[0084] **28** magnetic gap
[0085] **29** recess
[0086] **30** particle-optical component, e.g. steel tube
[0087] **31** quadrupole
[0088] **32** hexapole
[0089] **33**,**33'** current direction
[0090] **34** supply lines
[0091] **35** connection via windings on the rear side of the printed circuit board
[0092] **36** electron beam
[0093] **37** pole shoes
[0094] **38** windings
[0095] **39** iron shell
[0096] A,B,C,D,E,F through-connections

We claim:

1. Multipole coils comprising:
a first coil; and
at least one second coil, wherein said first and said second coils are disposed to concentrically enclose an imaginary axis, each of said first and said second coils having at least one winding disposed on a flexible printed circuit board via strip conductors, said printed circuit board being rolled into at least one printed circuit board layer.

2. The multipole coils of claim **1**, wherein each of said first and said second coils has windings, obtained through a spiral arrangement.

3. The multipole coils of claim **2**, wherein several windings per coil are generated through several printed circuit board layers, produced by rolling.

4. The multipole coils of claim **3**, wherein windings of coils of different printed circuit board layers extend in such a fashion that winding centers are always disposed along radii in a plane that extends perpendicularly to said imaginary axis.

5. The multipole coils of claim **1**, wherein windings form substantially rectangular windows.

6. The multipole coils of claim **5**, wherein long sides of said windows extend in an axial direction.

7. The multipole coils of claim **2**, wherein said windings are spirally arranged in several printed circuit board layers, outer spiral arrangements having more windings than inner ones.

8. The multipole coils of claim **1**, wherein a number of windings per printed circuit board layer is doubled by providing front and rear sides of said printed circuit board with strip conductors which form windings, wherein said strip conductors of printed circuit board layers are separated by an insulating layer.

9. The multipole coils of claim **3**, wherein different planes of strip conductors are interconnected by through-connections.

10. The multipole coils of claim **1**, wherein a cross-sectional surface of said strip conductors is increased through galvanic reinforcement.

11. The multipole coils of claim **3**, wherein adjustment markings on said printed circuit boards are used to obtain printed circuit board layers with exact positioning of said windings of respective coils.

12. The multipole coils of claim **11**, wherein said adjustment markings are bores through which a mandrel is inserted.

13. The multipole coils of claim **3**, wherein said printed circuit board layers are glued to each other.

14. The multipole coils of claim **3**, wherein thermally conducting materials are introduced for improving outward dissipation of heat.

15. The multipole coils of claim **1**, wherein additional electric or electronic components are disposed on said printed circuit board.

16. The multipole coils of claim **1**, wherein soft or hard magnetic materials are used to influence a magnetic flux in said printed circuit board.

17. The multipole coils of claim **1**, wherein the coils are structured to detect alternating magnetic fields.

18. The multipole coils of claim **1**, wherein the coils are structured for particle optics.

19. The multipole coils of claim **18**, wherein the coils are structured to be disposed around a vacuum tube.

20. The multipole coils of claim **18**, wherein the coils are used to correct aberrations.

21. The multipole coils of claim **18**, wherein the coils are structured as scanning coils in a scanning electron microscope.

22. The multipole coils of claim **1**, wherein winding centers of windings of each coil of a dipole are disposed on radii which form an angle of approximately 120° with respect to the imaginary axis.

23. The multipole coils of claim **1**, wherein several identical multipoles, different multipoles, dipoles, quadrupoles, hexapoles, or octupoles are disposed on said printed circuit board.

24. The multipole coils of claim **23**, wherein the multipoles have separate connections for current loads.

25. The multipole coils of claim **1**, wherein the coils comprise a first dipole and a second dipole which is rotated through 90° with respect to said first dipole.

26. The multipole coils of claim **1**, wherein several multipoles are distributed on several printed circuit board layers.

27. The multipole coils of claim **1**, wherein several multipoles are disposed on top of each other in an axial direction.

28. The multipole coils of claim **18**, wherein said printed circuit board is disposed in a beam passage opening of a lens.

29. The multipole coils of claim **28**, wherein the coils are disposed in a magnetic lens between a lens boundary and a magnetic gap.

30. The multipole coils of claim **18**, wherein said printed circuit board is inserted into a recess of a particle-optical component.

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